

- (a) Solve $\frac{x+1}{x} \geq 2$ 3
- (b) Find the acute angle between the lines $x + 3y = 4$ and $2x - 5y = 0$. Give your answer correct to the nearest degree. 3
- (c) If $\sqrt{3} \cos x - \sin x = R \cos(x + \theta)$, find the values of R and θ . 2
- (d) Evaluate $\int_0^1 \frac{2x dx}{(2x+1)^2}$, using the substitution $u = 2x + 1$. 4

Question 2. (Start a New Page)

- (a) It is given that $x^2 + x - 2$ is a factor of $x^3 + rx^2 - 4x + s$, where r and s are constants. 4
- (i) Show that $r + s = 3$.
- (ii) Evaluate r and s .
- (b) (i) What is the condition for the geometric series $a + ar + ar^2 + \dots$ to have a limiting sum? 4
- (ii) Consider the geometric series $1 - \tan^2 x + \tan^4 x + \dots$, where $0 < x < \frac{\pi}{2}$.
- For what values of x does this series have a limiting sum?
- (iii) Find the limiting sum in terms of $\cos x$.
- (c) Find the exact value of $\int_0^{\frac{\pi}{4}} \cos^2(\frac{1}{2}x) dx$. 4

Question 3 over the page

Question 3. (Start a New Page)

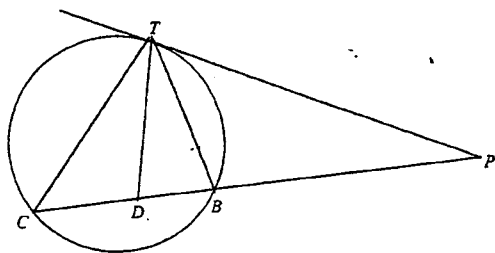
- | | Marks |
|---|-------|
| (a) (i) Sketch $y = 3 \sin x$ and $y = x$, for $0 \leq x \leq 2\pi$. | 4 |
| (ii) By substitution show that a solution for $3 \sin x - x = 0$ lies between $x = 2.2$ and $x = 2.4$. | |
| (iii) Taking $x = 2.3$ as an approximation to a solution of $3 \sin x - x = 0$, apply Newton's Method once to find a better approximation. Give your answer correct to 3 decimal places. | |
| (b) (i) Find $\frac{d}{dx} (2x \tan^{-1} x)$. | 4 |
| (ii) Hence, find the exact value of $\int_0^1 \tan^{-1} x dx$. | |
| (c) Use Mathematical Induction to show that, for all $n \geq 1$ | 4 |
| $1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + n \times 2^n = (n-1) \times 2^{(n+1)} + 2$ | |

Question 4 over the page

Question 4. (Start a New Page)

Marks

(a)



PT is a tangent to the circle and PBC is a secant. D is a point on PBC such that $TD = TB$.
Prove that $\angle CTD = \angle P$.

(b) Consider the function $f(x) = \frac{1}{1+x^2}$ for $x \leq 0$.

(i) Sketch $y = f(x)$. It is not necessary to show working.

(ii) Find the inverse function, $f^{-1}(x)$.

(iii) State the domain of $f^{-1}(x)$.

(c) (i) On the same set of axes sketch $y = \sin^{-1}x$ and $y = \cos^{-1}x$, showing all essential information.

(ii) Let $f(x) = \sin^{-1}x + \cos^{-1}x$.
By referring to the graph in part (i), or otherwise, explain why $f(x)$ is a constant function.

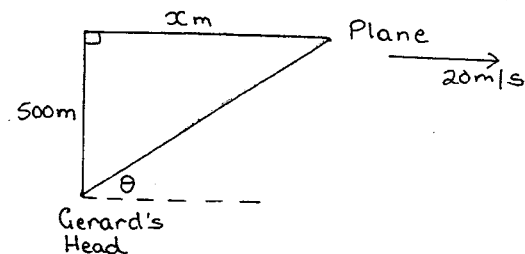
(iii) Hence, evaluate $\int_0^1 f(x) dx$.

Question 5 over the page

Question 5. (Start a New Page)

Marks
5

(a)



At 9 am an ultralight aircraft flies directly over Gerard's head, at a height of 500 metres. It maintains a constant speed of 20m/s, and a constant altitude.

If x is the horizontal distance travelled by the plane, and θ is the angle of elevation from Gerard's head to the plane,

(i) show that $\frac{dx}{d\theta} = -\frac{500}{\sin^2\theta}$.

(ii) Hence, show that $\frac{d\theta}{dt} = -\frac{1}{25} \sin^2\theta$.

(iii) find the rate of change of the angle of elevation at 9:01 am.

(b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x = 2at, y = at^2$.

(i) Find the co-ordinates of M , the midpoint of PQ .

(ii) Show that if the gradient of PQ is constant, the locus of M is a line parallel to the y -axis.

(d) (i) State the angle property of a cyclic quadrilateral.

(ii) Given that the quadrilateral $ABCD$ is cyclic, show that the sum of the tangents of the angles in the quadrilateral is zero.

That is:

$$\tan A + \tan B + \tan C + \tan D = 0.$$

Question 6. (Start a New Page)

(a) Find a general solution for x if $\tan x = \frac{1}{\sqrt{3}}$.

Give your answer in terms of π .

(b) (i) On the same set of axes graph $y = |2x - 1|$ and $y = 3x + 2$.

(ii) Hence, or otherwise, solve $|2x - 1| < 3x + 2$.

Question 6 continued over the page

Question 6. (Continued)

- (c) The rate at which a body cools is proportional to the difference between its temperature (T), and the constant temperature of the surrounding air (S).

That is $\frac{dT}{dt} = k(T - S)$, where t is the elapsed time and k is a constant.

- (i) Show that $T = S + Be^{kt}$, where B is a constant, is a solution of the above differential equation.
- (ii) A body cools from 150° to 90° in three hours. If the air temperature is 30°C , find the value of B and hence the value of k , correct to 3 decimal places.
- (iii) Using the values of B and k found in part (ii), determine the temperature of the body after a further three hours.

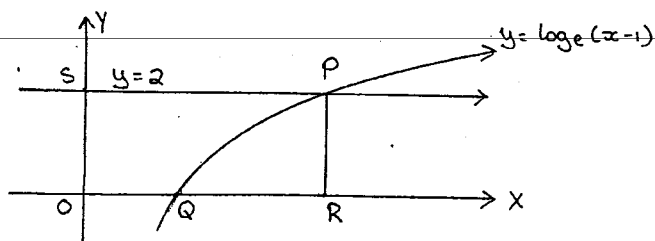
Question 7. (Start a New Page)

- (a) $P(x)$ is a polynomial of degree 3 with the following properties: $P(0) = 4$, $P(2) = 0$, $P(-2) = 0$ and $P(x)$ has a turning point at $x = -2$.

(i) Find $P(x)$.
(You may assume that $P(x) = ax^3 + bx^2 + cx + d$.)

- (ii) What is the nature of the turning point at $x = -2$?

- (b) The curve $y = \log_e(x - 1)$ meets the line $y = 2$ at the point P and the x -axis at the point Q . From P , perpendiculars are drawn to the x -axis and y -axis, meeting them at R and S , respectively, as shown in the diagram.



- (i) Show that the co-ordinates of P are $(e^2 + 1, 2)$.
- (ii) Show that the normal to the curve at Q passes through S .
- (iii) Show that the arc QP divides the rectangle $OSPR$ into two portions of equal area, where O is the origin.

End of Paper

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Question One. Start a new page

- (a) Evaluate to 4 significant figures

$$\frac{12 \times (1.05)^3}{2.31 \times 0.627}$$

- (b) Express in scientific notation, correct to 3 sig fig,

$$\sqrt[4]{\frac{4.3 \times 10^{18} - 2.9 \times 10^3}{2.4^3 + 3.31^2}}$$

- (c) Find the integers a and b such that

$$\frac{\sqrt{3}}{2 + \sqrt{3}} = a + b\sqrt{3}$$

- (d) Factorise $2ax + 4xb - a - 2b$.

- (e) The price of tickets to *Future World* has increased 5.5% to \$48. Find the price before the increase.

- (f) Solve and graph the solution on the number line

$$|6x - 9| > 21$$

Start a new page

the exact area enclosed between the curve $y = e^x$ and lines $x = 0$ and $x = 1$

the volume of the solid of revolution formed by rotating the graph $y = e^x$ and the lines $x = 0$ and $x = 1$ about the x -axis. Give your answer in terms of π .

equation of the parabola whose focus is $(-1, -2)$ and directrix is

He gives values for $f(x)$.

x	1	1.2	1.4	1.6	1.8
$f(x)$	1.7	1.8	1.9	2.0	2.2

Impson's rule to evaluate $\int_1^{1.8} f(x) \cdot dx$, correct to 2 decimal places.

all real numbers which satisfy the equation: $x^4 = 72 - x^2$

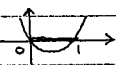
Marks

5

2

3

2

a) $x+1 > 2, x \neq 0$
 $\frac{x(x+1)}{x} > 2x^2$
 $x(x+1) > 2x^2$
 $-x^2 + x > 0$
 $-x(x-1) > 0$
 $x(x-1) < 0$

 Note $x \neq 0$
 $\therefore 0 < x < 1$

b) Line 1: $x+3y=4$
 $3y=4-x$
 $y=\frac{4-x}{3}$
 $\therefore m_1 = -\frac{1}{3}$

Line 2: $2x-5y=0$
 $5y=2x$
 $y=\frac{2}{5}x$
 $\therefore m_2 = \frac{2}{5}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{-\frac{1}{3} - \frac{2}{5}}{1 + (-\frac{1}{3})(\frac{2}{5})} \right|$
 $= \left| \frac{-\frac{5}{15} - \frac{6}{15}}{1 - \frac{2}{15}} \right|$
 $= \left| \frac{-\frac{11}{15}}{\frac{13}{15}} \right|$
 $= \frac{11}{13}$
 $\therefore \theta = 40.236 \dots$
 $\therefore$ acute angle is 40° (nearest degree)

c) $\sqrt{3} \cos \alpha - 2 \sin \alpha = R \cos(\alpha + \theta)$
 Since $R \cos(\alpha + \theta) = R \cos \alpha \cos \theta - R \sin \alpha \sin \theta$
 $R \cos \theta = \sqrt{3}$ and $R \sin \theta = 2$
 $\therefore \frac{R \sin \theta}{R \cos \theta} = \frac{2}{\sqrt{3}}$
 $\tan \theta = \frac{2}{\sqrt{3}}$
 $\theta = 30^\circ$

$R^2 \sin^2 \theta + R^2 \cos^2 \theta = 1 + 3$
 $R^2 = 4$
 $R = 2$

OR
 $R = \sqrt{(\sqrt{3})^2 + 2^2}$
 $= \sqrt{7}$
 $= 2$

$\therefore \sqrt{3} \cos \alpha - 2 \sin \alpha = 2 \cos(\alpha + \theta)$
 $\frac{\sqrt{3}}{2} \cos \alpha - \frac{1}{2} \sin \alpha = \cos \alpha \cos \theta - \sin \alpha \sin \theta$
 $\therefore \cos \theta = \frac{\sqrt{3}}{2}$ and $\sin \theta = \frac{1}{2}$
 $\Rightarrow \theta = 30^\circ$

d) $\int \frac{2x \, dx}{(2x+1)^2}$ $u = 2x+1$
 $\frac{du}{dx} = 2$
 $\frac{du}{2} = dx$
 When $x=0, u=1$
 $x=1, u=3$
 Also $2x = u-1$

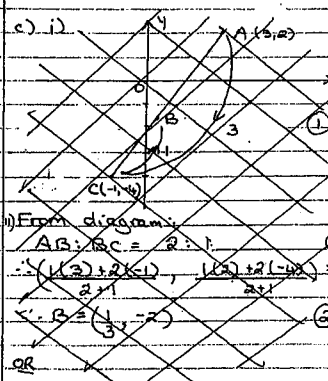
Q1 d) cont)
 $\therefore \int \frac{u-1}{u^2} \frac{du}{2}$
 $= \frac{1}{2} \int \left(\frac{u}{u^2} - \frac{1}{u^2} \right) du$
 $= \frac{1}{2} \int \left(\frac{1}{u} - \frac{1}{u^2} \right) du$
 $= \frac{1}{2} \left[\ln u + \frac{1}{u} \right]$
 $= \frac{1}{2} \left[(\ln 3 + \frac{1}{3}) - (\ln 1 + 1) \right]$
 $= \frac{1}{2} (\ln 3 + \frac{1}{3} - 0 - 1)$
 $= \frac{1}{2} (\ln 3 - \frac{2}{3})$

Question 2
 a) Let $P(x) = x^2 + rx^2 - 4x + 0$
 also $x^2 + x - 2 = (x+2)(x-1)$
 Since $x^2 + x - 2$ is a factor of $P(x)$
 $P(-2) = 0$ and $P(1) = 0$
 i) Since $P(1) = 0$
 $1 + r(1)^2 - 4(1) + 0 = 0$
 $r + 0 = 3$
 $r = 3$
 ii) Since $P(-2) = 0$
 $(-2)^2 + r(-2)^2 - 4(-2) + 0 = 0$
 $4 + 4r + 8 + 0 = 0$
 $4r + 12 = 0$
 $r = -3$
 Sub. (B) into (A)
 $r + (-4r) = 3$
 $-3r = 3$
 $r = -1$
 $\therefore r = 4$

b) i) A geometric series
 $a, ar, ar^2, \dots$ has a limiting sum when $|r| < 1$

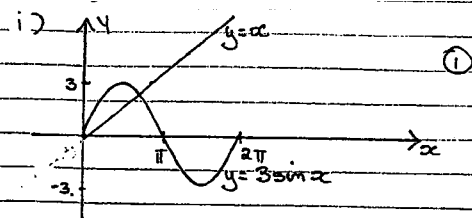
ii) For the series
 $1 - \tan^2 x + \tan^4 x - \dots$
 $r = -\tan^2 x$
 $\therefore$ if $|r| < 1$
 $1 - \tan^2 x < 1$
 $\tan^2 x < 1$
 But $\tan^2 x \geq 0$ and $0 < x < \frac{\pi}{2}$
 $\therefore 0 < x < \frac{\pi}{4}$

iii) $S_\infty = \frac{a}{1-r}$
 $= \frac{1}{1 + \tan^2 x}$
 $= \frac{1}{\sec^2 x}$
 $= \cos^2 x$

c) i) 
 From diagram
 $AB:BC = 2:1$
 $\therefore \frac{1(2) + 2(-1)}{2+1}, \frac{1(2) + 2(-1)}{2+1} = 0$
 $\therefore B = (1, -2)$
 OR

$AC:CB = 3:-1$
 $A(3,2) \quad C(-1,-4)$
 $m:n = 3:-1$
 $\therefore B = (x,y)$
 Then
 $-1 = \frac{-1 \times 3 - 3 \times x}{3-1}$ ①
 $-2 = \frac{-1 \times 2 + 3 \times y}{3-1}$
 $\frac{1}{3} = x$
 and
 $-4 = \frac{-1 \times 2 + 3 \times y}{3-1}$
 $-8 = 3y - 2$
 $-6 = 3y$
 $-2 = y$
 $\therefore B$ is $(\frac{1}{3}, -2)$

c) $\int_0^{\frac{\pi}{4}} \cos^2(\frac{1}{2}x) dx$
 Since $\cos^2 A = \frac{1}{2}(\cos 2A + 1)$
 $\cos^2 \frac{1}{2}x = \frac{1}{2}(\cos x + 1)$ ①
 $\therefore \int_0^{\frac{\pi}{4}} \cos^2(\frac{1}{2}x) dx$
 $= \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 + \cos x) dx$
 $= \frac{1}{2} [x + \sin x]_0^{\frac{\pi}{4}}$ ①
 $= \frac{1}{2} (\frac{\pi}{4} + \sin \frac{\pi}{4})$ ①
 $= \frac{1}{2} (\frac{\pi}{4} + \frac{1}{\sqrt{2}})$ ①



ii) When $x = 2.2$
 $3 \sin x - x = 3 \sin 2.2 - 2.2$
 $= 0.2254...$
 When $x = 2.4$
 $3 \sin x - x = 3 \sin 2.4 - 2.4$
 $= -0.3736...$
 Since the sign of $3 \sin x - x$ changes from $x = 2.2$ to $x = 2.4$, then the solution lies between $x = 2.2$ and $x = 2.4$.

iii) $f(x) = 3 \sin x - x$
 $f'(x) = 3 \cos x - 1$
 $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $= 2.3 - \frac{3 \sin(2.3) - (2.3)}{3 \cos(2.3) - 1}$ ①
 $= 2.27903...$
 $= 2.279$ (3 dp) ①

b) i) $\frac{d}{dx} (2x \tan^{-1} x)$
 $= (\tan^{-1} x) \times 2 + 2x \times \frac{1}{1+x^2}$
 $= 2 \tan^{-1} x + \frac{2x}{1+x^2}$ ①

$\therefore 2 \tan^{-1} x = \frac{d}{dx} (2x \tan^{-1} x) - \frac{2x}{1+x^2}$

Q3(b) (cont)

ii) $\int_0^1 \tan^{-1} x dx$
 $= \int_0^1 \frac{d}{dx} (2x \tan^{-1} x) dx$
 $= \int_0^1 \frac{2x}{1+x^2} dx$ ①
 $= [2x \tan^{-1} x]_0^1$
 $= [\log_e (1+x^2)]_0^1$ ①
 $= [2 \tan^{-1} 1 - 0]$
 $= [\log_e 2 - \log_e 1]$
 $= 2 \times \frac{\pi}{4} - \log_e 2$
 $\therefore \int_0^1 \tan^{-1} x dx$
 $= \frac{\pi}{4} - \frac{1}{2} \log_e 2$ ①

c) Step 1: Let $n=1$
 LHS = $1 \times 2 = 2$
 RHS = $(1-1) \times 2^{2(1)} + 2$
 $= 2$
 $\therefore$ true when $n=1$ ①

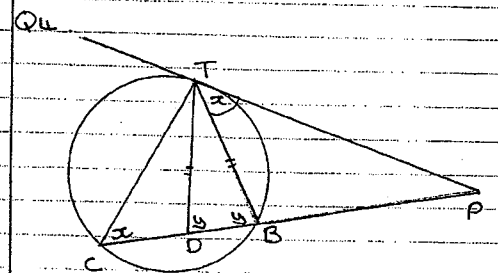
Step 2: Assume formula true when $n=k$
 i.e.
 $1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k$
 $= (k-1) \times 2^{k+1} + 2$

Step 3: When $n=k+1$
 $1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k + (k+1) \times 2^{k+1}$
 $= (k-1) \times 2^{k+1} + 2 + (k+1) \times 2^{k+1}$ ①

Q3(c) (cont)
 $= 2^{k+1} (k-1+k+1) + 2$
 $= 2k \times 2^{k+1} + 2$
 $= k \times 2^{k+2} + 2$
 $= (k+1-1) \times 2^{(k+1)+1} + 2$ ①

$\therefore$ If true when $n=k$, then formula true when $n=k+1$

Step 4: But, formula is true when $n=1 \therefore$ true when $n=1+1$ or $n=2 \therefore$ true when $n=2+1$ or $n=3$ etc.
 $\therefore$ Formula is true for all n ①



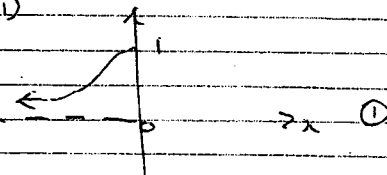
Let $\angle TCB = x^\circ$
 In $\triangle TDB$, $\angle O = \angle B = y^\circ$ ①
 (base angles of isosceles $\triangle$)

Also $\angle PTB = \angle TCB$
 $= x^\circ$ ①
 (Alternate segment theorem)

In $\triangle TCD$
 $\angle T + \angle C = \angle TDP$ (ext. $\angle$ of $\triangle$ ① Theorem)

$\therefore \angle CTD = y - x^\circ$
 Similarly in $\triangle TBP$, $\angle P = y - x^\circ$
 $\therefore \angle CTD = \angle P$ ①

Q4 (cont)
b) i)



ii) For inverse:

$$x = \frac{1}{1+y^2}, y \leq 0$$

$$1+y^2 = \frac{1}{x} \quad (1)$$

$$y^2 = \frac{1}{x} - 1$$

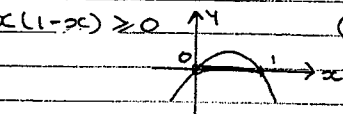
$$y = \pm \sqrt{\frac{1}{x} - 1}, y \leq 0$$

$$\therefore y = -\sqrt{\frac{1-x}{x}} \quad (1)$$

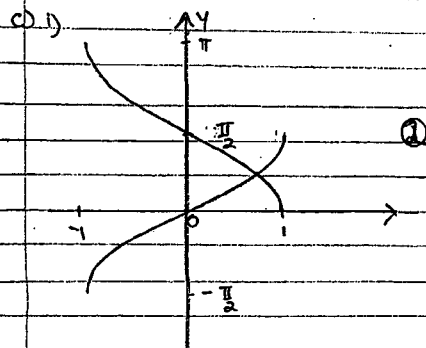
iii) Domain of inverse function

$$\frac{1-x}{x} \geq 0 \text{ and } x \neq 0$$

$$x(1-x) \geq 0 \quad (1)$$



$$0 < x \leq 1$$



Q4 c) (cont)

ii) By adding ordinates at some key points on the graph and by noting the symmetry of the graphs it can be seen that

$$f(x) = \sin^{-1}x + \cos^{-1}x = \text{constant} \quad (1)$$

$$(\therefore \frac{\pi}{2})$$

$$\text{OR } f(x) = \sin^{-1}x + \cos^{-1}x$$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$$

$$\text{Note } f(0) = \sin^{-1}(0) + \cos^{-1}(0) = \frac{\pi}{2}$$

$$\text{iii) } \int_0^1 f(x) dx = \int_0^1 \frac{\pi}{2} dx = \left[\frac{\pi}{2} x \right]_0^1 = \frac{\pi}{2}$$

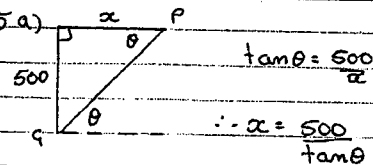
OR

$$\text{From the graph: } \int_0^1 (\sin^{-1}x + \cos^{-1}x) dx$$

= area of rectangle with width 1 and height $\frac{\pi}{2}$

$$\therefore \text{Area} = 1 \times \frac{\pi}{2} = \frac{\pi}{2}$$

Q5 a)



$$\tan \theta = \frac{500}{1200}$$

$$\therefore \alpha = \frac{500}{\tan \theta}$$

$$\text{i) } \frac{d\alpha}{d\theta} = \frac{(\tan \theta) 0 - 500 \sec^2 \theta}{\tan^2 \theta} \quad (1)$$

$$= \frac{-500 \times \cos^2 \theta}{\cos^4 \theta \times \frac{1}{\sin^2 \theta}}$$

$$= \frac{-500}{\sin^2 \theta} \quad (1)$$

$$(\therefore -500 \operatorname{cosec}^2 \theta)$$

$$\text{ii) } \frac{d\alpha}{dt} = 20$$

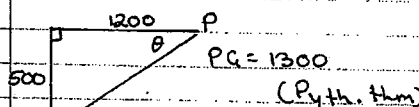
$$\frac{d\theta}{dt} = \frac{d\theta}{d\alpha} \times \frac{d\alpha}{dt}$$

$$= \frac{1}{-500 \operatorname{cosec}^2 \theta} \times 20$$

$$= -\frac{1}{25} \sin^2 \theta \quad (1)$$

$$\text{iii) At 10.1 am, } t = 60$$

$$x = 20 \times 60 = 1200$$



$$\therefore \sin \theta = \frac{500}{1300} \quad (1)$$

$$\therefore \frac{d\theta}{dt} = -\frac{1}{25} \times \sin^2 \theta$$

$$= -\frac{1}{25} \times \left(\frac{5}{13}\right)^2$$

$$= -\frac{1}{169} \text{ degrees/sec} \quad (1)$$

b) $P(2ap, ap^2)$ $Q(2aq, aq^2)$

$$\text{i) } M = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right) = (a(p+q), a \frac{p^2+q^2}{2})$$

ii) let $m = \text{Grad. of PQ}$

$$m = \frac{aq^2 - ap^2}{2aq - 2ap}$$

$$= \frac{q^2 - p^2}{2(q-p)}$$

$$= \frac{(q-p)(q+p)}{2(q-p)}$$

$$= \frac{q+p}{2} \quad (1)$$

Now if m is constant, then $\frac{q+p}{2} = k$

$$\text{or } q+p = 2k \quad (1)$$

$\therefore x$ -co-ord of midpoint, M , is $x = a(p+q)$

$$= 2ak$$

$$= \text{constant} \quad (1)$$

$\therefore$ Locus of M is a line parallel to the y -axis.

c) Possible outcomes

(1,5)	(2,5)	(3,5)	(4,5)	(5,5)
(6,5)	(5,4)	(5,3)	(5,2)	(5,1)
(4,4)	(4,3)	(4,2)	(4,1)	(3,4)
(3,3)	(3,2)	(3,1)	(2,3)	(2,2)
(2,1)	(1,2)	(1,1)	(1,4)	(1,3)

Sample space = 16

Favourable outcomes = 2

$$P(\text{both's 8}) = \frac{2}{16}$$

Q5 d)

Since ABCD is a cyclic quad.

$$\angle A + \angle C = 180 \quad (1)$$

$$\therefore \angle C = 180 - \angle A$$

$$\Rightarrow \tan C = -\tan A \quad (1)$$

Similarly $\tan D = -\tan B$

$$\begin{aligned} \therefore \tan A + \tan B + \tan C + \tan D \\ = \tan A + \tan B - \tan A - \tan B \\ = 0 \end{aligned}$$

Q6

$$a) \tan x = \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6}, \pi + \frac{\pi}{6} \quad (1)$$

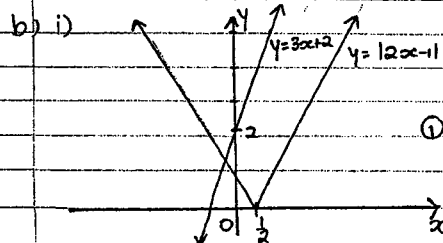
for $0 \leq x < 2\pi$

General solution:

$$x = \frac{\pi}{6} \pm 2n\pi, \pi + \frac{\pi}{6} \pm 2n\pi$$

$$= \frac{\pi}{6} \pm n\pi \text{ for all } (1)$$

integral n.



ii) Pt. of intersection:

$$y = 3x + 2 \text{ and } y = 1 - 2x$$

$$3x + 2 = 1 - 2x$$

$$5x = -1$$

$$x = -\frac{1}{5} \quad (1)$$

Q6 b) ii) (cont)

$\therefore$ From the graph

$$|2x-1| < 3x+2 \text{ for } x > -\frac{1}{5} \quad (1)$$

$$\text{OR } |2x-1| < 3x+2$$

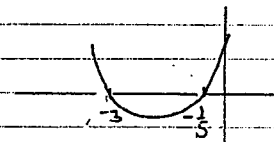
$$(2x-1)^2 < (3x+2)^2$$

$$(2x-1)^2 - (3x+2)^2 < 0$$

$$(2x-1+3x+2)(2x-1-(3x+2)) < 0$$

$$(5x+1)(-x-3) < 0$$

$$(5x+1)(x+3) > 0$$



$$\therefore x < -3, x > -\frac{1}{5}$$

BUT: when $x < -3$, $3x+2 < 0$

$\therefore x > -\frac{1}{5}$ is only soln.

c) i)

$$T = S + Be^{kt} \quad (A)$$

$$\frac{dT}{dt} = 0 + B \times k e^{kt}$$

$$\frac{dT}{dt} = B k e^{kt} \quad (1)$$

but: $Be^{kt} = T - S$ from (A)

$$\therefore \frac{dT}{dt} = k(T - S) \quad (1)$$

$\therefore T = S + Be^{kt}$ is a solution

ii)

Q6 c) (cont)

ii) When $t=0$ $T=150$

$$S = 30$$

$$T = S + Be^{kt}$$

$$150 = 30 + Be^0$$

$$\therefore B = 120 \quad (1)$$

When $t=3$ $T=90$

$$\therefore 90 = 30 + 120e^{3k}$$

$$60 = 120e^{3k}$$

$$0.5 = e^{3k}$$

$$k = \frac{1}{3} \log_e 0.5 \quad (1)$$

$$= -0.231 \quad (3 \text{ dp})$$

iii) When $t=6$

$$T = 30^\circ + 120e^{6 \times \frac{1}{3} \log_e 0.5}$$

$$= 30^\circ + 120e^2$$

$$= 30 + 30$$

$$= 60 \quad (1)$$

$\therefore$ Temperature is 60° after a further 3 hours.

Q7 a) $P(x) = ax^3 + bx^2 + cx + d$

$$i) P(0) = 4$$

$$\therefore d = 4$$

$\Rightarrow$ y-intercept is 4 (1)

$$P(2) = 0$$

$$\therefore 8a + 4b + 2c + 4 = 0 \quad (A)$$

$$4a + 2b + c + 2 = 0$$

$$P(-2) = 0$$

$$-8a + 4b - 2c + 4 = 0 \quad (B)$$

$$-4a + 2b - c + 2 = 0$$

Q7 a) (cont)

Since $x = -2$ is a turning point,

$$P'(-2) = 0$$

$$P'(x) = 3ax^2 + 2bx + c$$

$$\therefore 12a - 4b + c = 0 \quad (C)$$

$$3A - C \quad 12a + 6b + 3c + 6 = 0$$

$$12a - 4b + c = 0$$

$$10b + 2c + 6 = 0$$

$$5b + c + 3 = 0 \quad (D)$$

$$C + 3D \quad 12a - 4b + c = 0$$

$$-12a + 6b - 3c + 6 = 0$$

$$2b - 2c + 6 = 0$$

$$b - c + 3 = 0 \quad (E)$$

$$D - E$$

$$5b + c + 3 = 0$$

$$b = -1$$

$$b + c + 3 = 0$$

$$c = 2$$

$$4b = 0$$

$$b = 0 \quad (1)$$

$$\therefore c = -3 \quad (1)$$

Sub. into (C)

$$12a - 0 + 3 = 0 \quad a = -\frac{1}{4}$$

$$12a = -3$$

$$a = -\frac{1}{4} \quad (1)$$

$$\therefore P(x) = -\frac{x^3}{4} - 3x + 4 = 0$$

$\therefore$ Graph looks like $= \frac{1}{4}(x-2)(x+2)$



$\therefore$ Turning point at $x = -2$ is a relative minimum (1)

(18)

Q7 b)

i) At P $y=2$

$$\therefore \log_e(x-1) = 2$$

$$x-1 = e^2$$

$$x = e^2 + 1 \quad (1)$$

$\therefore P$ is $(e^2 + 1, 2)$

ii) $y = \log_e(x-1)$

$$\frac{dy}{dx} = \frac{1}{x-1}$$

at Q $x=2$, $y=0$

$$\therefore \frac{dy}{dx} = \frac{1}{2-1} = 1$$

$\therefore$ gradient of normal is $m = -1$ (1)

Eqn. of normal:

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -1(x - 2)$$

$$y = 2 - x \quad (1)$$

At S $x=0$, $y=2$

$\therefore$ on the line $y = 2 - x$

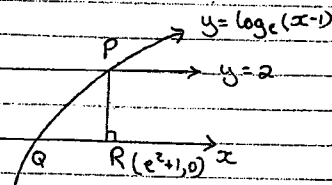
when $x=0$, $y=2$

$\therefore S$ lies on the normal at Q (1)

iii)

S(0,2)

O



Area of OSPR

$$= (e^2 + 1) \times 2$$

$$= 2(e^2 + 1) \quad (1)$$

Area OSPQ

$$= \int_0^2 g(y) dy$$

$$= \int_0^2 (e^y + 1) dy \quad (1)$$

$$= [e^y + y]_0^2$$

$$= (e^2 + 2) - (e^0 + 0)$$

$$= e^2 + 1 \quad (1/2)$$

$\therefore$ Area of OSPQ = $\frac{1}{2}$ area of rectangle OSPR (1/2)